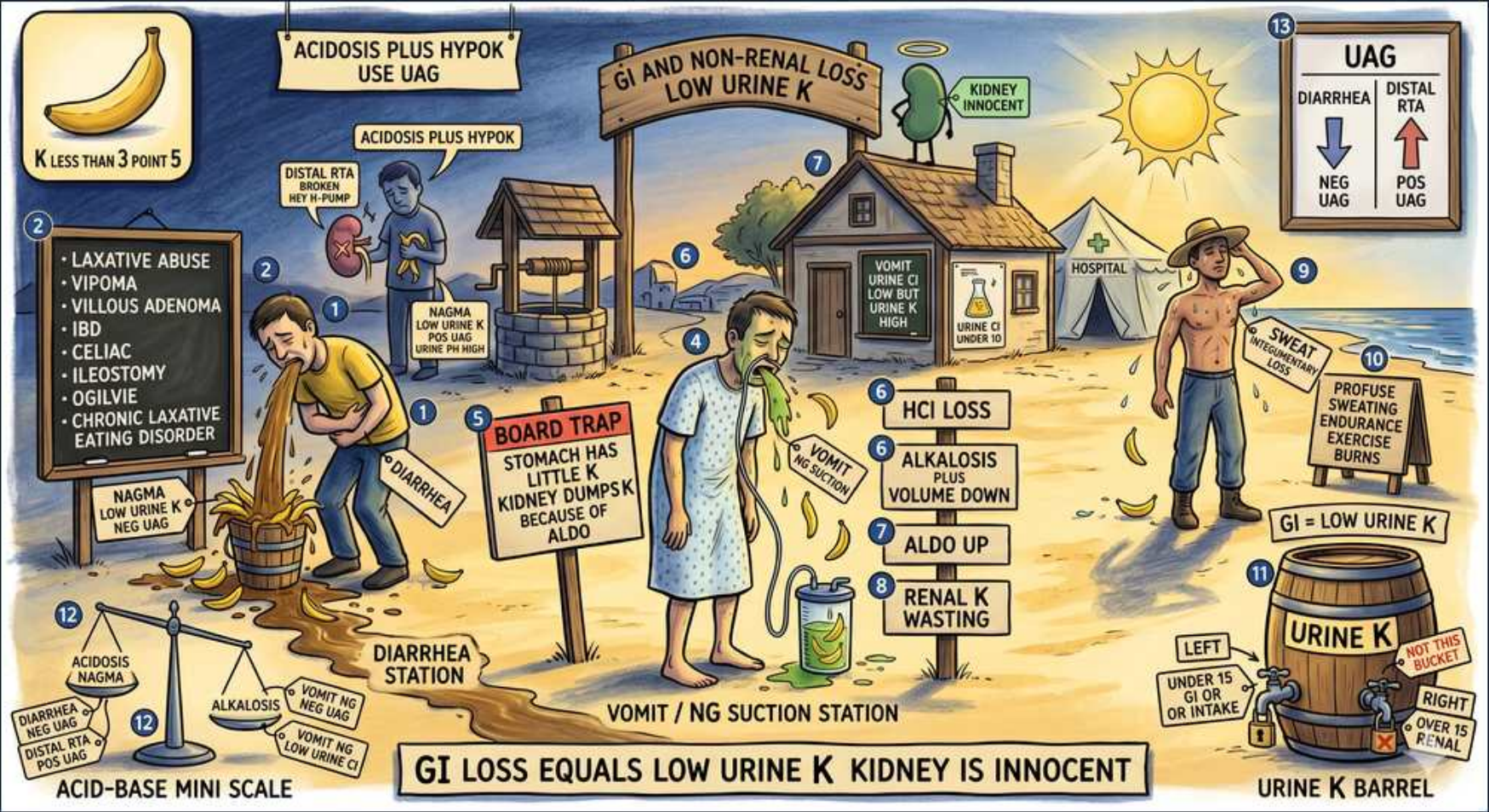


Hypokalemia — GI / extrarenal loss



1. GI loss = LOW urine K

WHAT YOU SEE

Banner: GI and non-renal loss, low urine K, kidney innocent

WHAT IT ENCODES

Extrarenal (GI or skin) K loss → the kidney appropriately conserves K → spot urine K <15 mmol/day (or <13 mmol/g Cr, TTKG <3). The kidney is NOT the leak; it is doing its job. This is the LEFT, low-urine-K arm of the hypokalemia algorithm.

BOARD TRAP

Ordering renin/aldosterone first when urine K is low. Low urine K rules OUT a renal/diuretic cause, so the mineralocorticoid workup is wasted — the next step is acid-base status and the urine anion gap, not aldosterone.

2. Diarrhea causes

WHAT YOU SEE

List: laxative abuse, VIPoma, villous adenoma, IBD, celiac, ileostomy, Ogilvie, chronic laxative, eating disorder

WHAT IT ENCODES

Lower-GI (secretory/diarrheal) K losses: laxative abuse, VIPoma (WDHA — Watery Diarrhea, Hypokalemia, Achlorhydria), villous adenoma, IBD, celiac, high-output ileostomy, Ogilvie. Stool is rich in K and HCO₃ → hypokalemia WITH a non-anion-gap (hyperchloremic) metabolic ACIDOSIS.

BOARD TRAP

VIPoma is the secretory-diarrhea zebra: profound hypokalemia plus tea-colored watery diarrhea that persists with FASTING (osmotic/laxative diarrhea stops when fasting). Achlorhydria + flushing point to VIP, not laxatives.

3. Diarrhea man

WHAT YOU SEE

Patient labeled #1 with diarrhea at the well

WHAT IT ENCODES

Diarrhea produces volume contraction with normal or LOW blood pressure plus low urine K — the non-renal route confirmed clinically. The accompanying acidosis is hyperchloremic (normal anion gap) from stool bicarbonate loss.

BOARD TRAP

Expecting hypertension. A normotensive, volume-deplete hypokalemic patient with acidosis is GI/RTA territory; HTN would redirect you to mineralocorticoid excess instead.

4. Diarrhea → acidosis + neg UAG

WHAT YOU SEE

Sign: acidosis plus hypoK, use UAG; distal RTA growth higher up

WHAT IT ENCODES

Stool HCO₃ loss drives a non-anion-gap metabolic acidosis. The kidney responds correctly by excreting NH₄Cl, so the urine anion gap (Na + K – Cl) is NEGATIVE (NH₄⁺ unmeasured, raising urine Cl). Negative UAG = extrarenal/GI source.

BOARD TRAP

Distal (type 1) RTA gives a POSITIVE urine anion gap because the kidney CANNOT excrete NH₄⁺. Same hypokalemia + non-gap acidosis picture, opposite UAG — the sign is the discriminator, and confusing them is the classic trap.

5. Board trap: low urine K from aldo

WHAT YOU SEE

Red BOARD TRAP sign: stomach has little K, kidney dumps K because of aldo

WHAT IT ENCODES

In VOMITING, gastric fluid contains little K (~5–10 mmol/L). The real K loss is RENAL: loss of HCl → metabolic alkalosis, the bicarbonaturia obligates Na/K excretion, and volume depletion → secondary hyperaldosteronism → distal K wasting via ROMK.

BOARD TRAP

Classifying vomiting as a 'GI/low-urine-K' cause. Mechanistically it is RENAL K loss, and EARLY urine K can be inappropriately HIGH; use urine CHLORIDE (low, <20) — not urine K or Na — to recognize the volume-deplete, saline-responsive alkalosis.

6. Vomiting man

WHAT YOU SEE

Patient #5 vomiting green at vomit/NG suction station

WHAT IT ENCODES

Upper-GI loss (vomiting or NG suction) loses H⁺ and Cl⁻ → hypochloremic metabolic ALKALOSIS with volume depletion. Same mechanism whether self-induced (bulimia) or iatrogenic (NG tube).

BOARD TRAP

Suspected bulimia is confirmed by metabolic alkalosis + low urine Cl + LOW urine K once volume-replete; finding HIGH urine Cl instead points away from vomiting toward a diuretic or Bartter/Gitelman.

7. HCl loss → alkalosis

WHAT YOU SEE

Signs #6: HCl loss; alkalosis plus volume down; aldo up; renal K wasting

WHAT IT ENCODES

Vomiting cascade: lose H⁺/Cl⁻ → metabolic alkalosis + hypovolemia → secondary hyperaldosteronism → renal K AND H⁺ wasting (paradoxical aciduria). Treatment is normal saline (saline-responsive alkalosis) plus KCl.

BOARD TRAP

Trying to correct the alkalosis with more chloride-free fluid or acetazolamide first. The fix is volume/chloride repletion with NS; failure to replace Cl perpetuates the alkalosis and the renal K leak.

8. Sweating / skin loss

WHAT YOU SEE

Cowboy #10 sweating in the sun: profuse sweating, endurance exercise, burns

WHAT IT ENCODES

Skin is the second extrarenal route: profuse sweating, endurance exercise in heat, and extensive burns lose K (and Na) → low urine K, like GI loss. Usually modest unless losses are large and sustained.

BOARD TRAP

Overlooking burns/heavy sweat as a cause and labeling it 'idiopathic.' A low urine K in an athlete or burn patient localizes the loss to skin — no renal workup needed.

9. Urine K barrel

WHAT YOU SEE

Barrel labeled URINE K, left under-15 conserving (correct), right high spilling

WHAT IT ENCODES

Spot/24-hr urine K is the pivot test of the whole algorithm. Extrarenal loss → urine K <15 mmol/day (kidney conserving correctly). Renal loss → >15 mmol/day (inappropriately high). TTKG <3 supports extrarenal, >4 supports renal.

BOARD TRAP

Reading urine K in isolation. Always pair it with acid-base status AND blood pressure — e.g. vomiting can show transiently high urine K despite being volume-deplete, so urine Cl must back it up.

10. UAG box

WHAT YOU SEE

UAG panel: diarrhea = negative UAG, distal RTA = positive UAG

WHAT IT ENCODES

Urine anion gap = (urine Na + urine K) – urine Cl, used as a surrogate for unmeasured NH₄⁺. NEGATIVE UAG = appropriate NH₄Cl excretion = GI/diarrhea. POSITIVE UAG = failure to excrete NH₄⁺ = distal (type 1) RTA. Memory aid: 'neGUTive' = gut cause.

BOARD TRAP

Applying UAG when there's an anion-gap acidosis or massive bicarbonaturia — it is only valid in a non-anion-gap (hyperchloremic) acidosis. Misapplying it to a ketoacidosis or to vomiting gives a meaningless number.

11. Acid-base mini scale

WHAT YOU SEE

Mini balance: acidosis = nagma/neg UAG, alkalosis side

WHAT IT ENCODES

Acid-base status frames the differential: hypokalemia + non-gap metabolic ACIDOSIS → diarrhea (neg UAG) or RTA (pos UAG). Hypokalemia + metabolic ALKALOSIS → vomiting/NG suction, diuretics, or mineralocorticoid excess.

BOARD TRAP

Pairing diarrhea with alkalosis (or vomiting with acidosis). Diarrhea = acidosis, vomiting = alkalosis — flipping these is a frequently tested error.

12. Acidosis + hypoK → use UAG

WHAT YOU SEE

Top-left banner: acidosis plus hypoK use UAG

WHAT IT ENCODES

Workflow anchor: when hypokalemia comes with a non-gap acidosis, the NEXT step is the urine anion gap (to separate diarrhea from RTA), not renin/aldosterone. Reserve the mineralocorticoid workup for the hypertensive, alkalotic patient.

BOARD TRAP

Jumping to imaging the adrenals in a hypokalemic-acidotic patient. Acidosis takes you to the UAG/RTA pathway; adrenal/renin-aldo workup belongs to the hypertensive-alkalosis arm only.